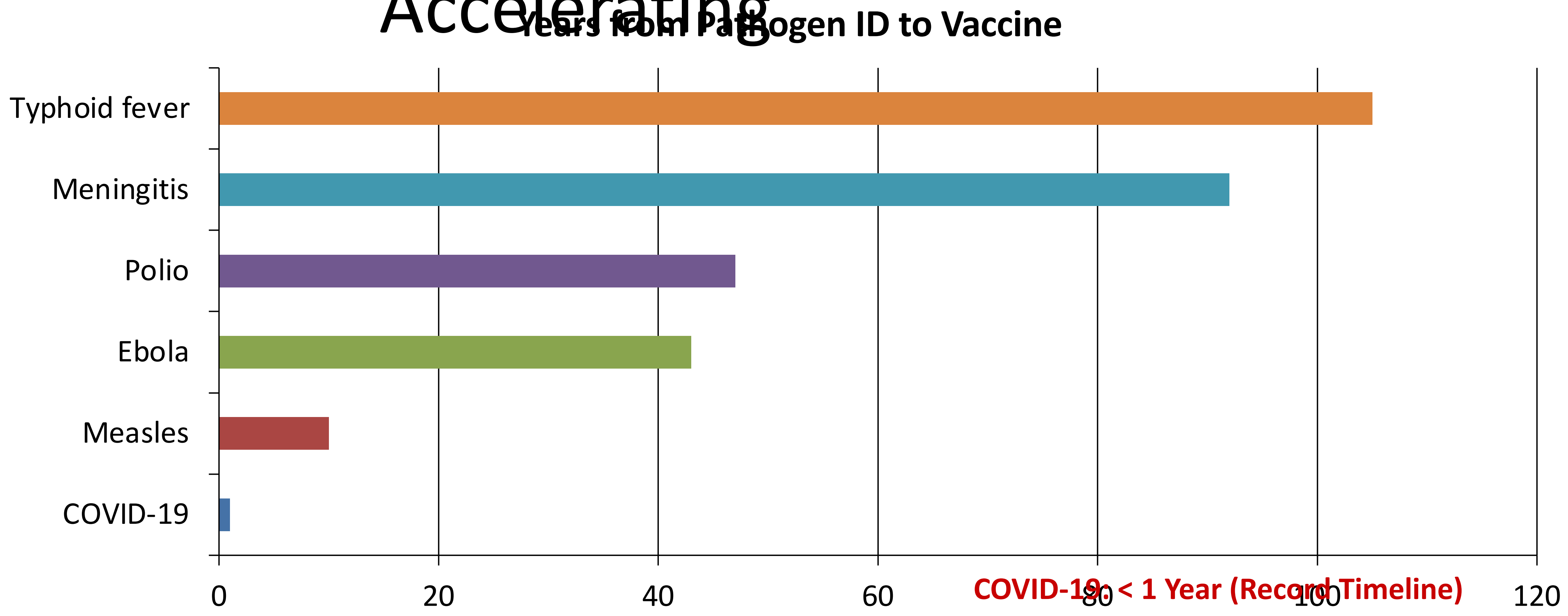
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

1796–2020: A Data-Driven Overview\nSource: Our World in Data

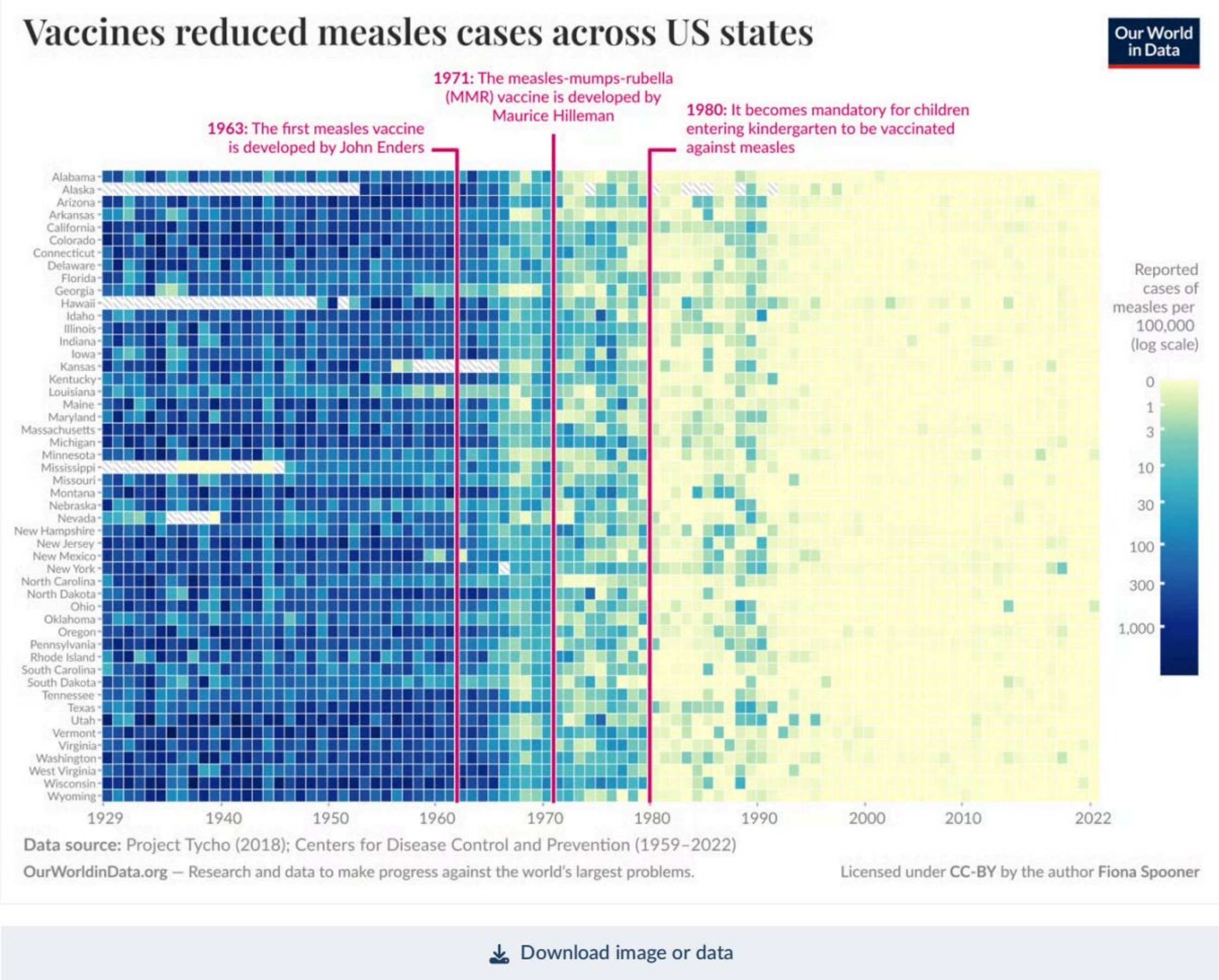
From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in

Disease	Pre-Vaccine Cases	Post-Vaccine Cases	Case Reduction	Pre-Vaccine Deaths	Death Reduction
Diphtheria	158	0	**100%**	13.7	**100%**
Measles	3,044	0.2	99.99%	2.5	**100%**
Smallpox	250	0	**100%**	2.9	**100%**
Acute polio	141	0	**100%**	10	**100%**
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	**100%**
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Measles Cases Collapsed Across All 50 US States After Vaccine



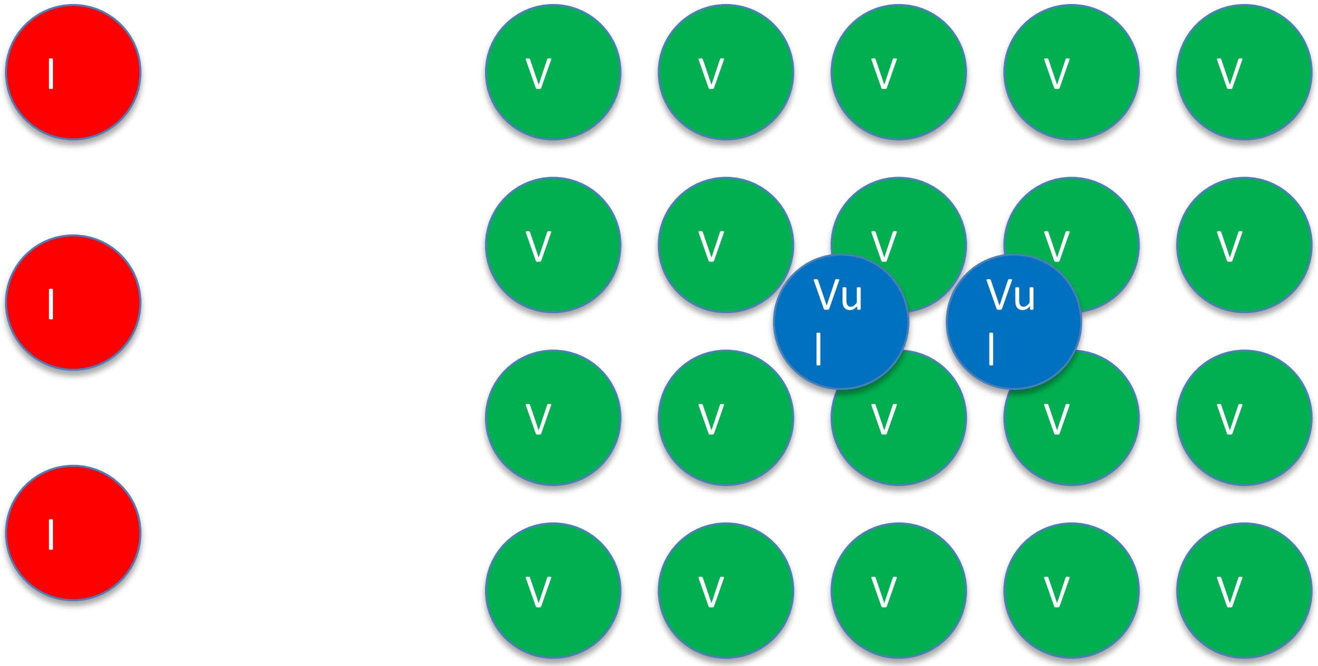
Key Milestones:

- 1963: First measles vaccine developed
- 1971: MMR vaccine developed
- 1980: Vaccination mandatory for kinder

Smallpox: The Only Human Disease Eradicated Through Vaccination

- Declared eradicated worldwide in 1980 — a singular achievement.
- **Timeline:**
  - 1959: WHO passes resolution to eradicate smallpox
  - 1967: WHO launches intensified eradication campaign
  - 1977: Last natural case recorded in Somalia
  - 1980: Global eradication certified
- **Success Factors:**

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Legend:

Green (V): Vaccinated (Shielding)

Blue (Vul): Vulnerable (Cannot be vaccinat

Red (I): Infected (Pathogen blocked)

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

- While many diseases are eliminated, significant challenges remain:
  - Pneumococcal disease: Only 40.5% case reduction and 31.3% death reduction.
  - Hepatitis A: 89% case reduction, indicating incomplete control.
  - Hepatitis B: 83.9% case reduction, requiring better coverage.
- **Key Barriers:**
  - Global access and equity gaps
  - Supply shortages putting children at risk

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the US.
- Vaccine development timelines have collapsed from over 100 years to under 1 year (COVID-19).
- Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.
- Herd immunity shields vulnerable